Surgical Robotics & Navigation · Verified · Handed over

rTMS Coil Navigation Software (NeuroPilot)

Turn optical tracking and 3D Slicer into a license-gated coil-navigation product that gives operators position and angle guidance they can rely on.

Period 2024.07 - 2026.08 | Evidence state Verified · Completed (handed over 2026.08) | Portfolio tier Surgical Robotics & Navigation

Designed the 3D Slicer workflow from procedure setup to real-time coil navigation, the landmark-and-ICP registration engine, tracker, tablet, and robot integration, and a license-gated product structure, delivered it to the customer, and handed it over to the customer's development team in August 2026, after which I take part as an adviser.



Figure 1. Demonstration of the TMS coil-navigation robot system on an exhibition stand: robot arm, treatment chair, coil, and navigation displays in one setup.

Problem and system boundary

Medical images, a tracker, a tablet, and a robot each sit in their own coordinate frame and had to become one verifiable flow.

Owned decisions and implementation

My role

Led the 3D Slicer procedure-setup and navigation workflow, the registration engine combining landmarks and ICP with its coordinate-transform chain, the optical-tracker SDK migration with the tablet protocol and robot integration, and the product structure covering license verification and optional-feature gating. From June to August 2026 I migrated the optical-tracker SDK from v6 to v7.1.2, merging five core DLLs into a single DtSkadi.dll, and built a product-license scheme with product and option gating, PC binding, and an issuing tool. Customer-provided AI modules (landmarks, treatment position, coil-angle optimization) connect through an adapter boundary that is updated by swapping a folder, with an angle-optimization UI, progress display and cancellation, project reset, and multiple-instance prevention. The tablet TCP protocol is a port of the C++ NaviClient protocol a colleague wrote. For a health-technology R&D programme, also owned the registration workflow, result panel and matrix export, and tracker-SDK integration improvements of a depth-camera (LiDAR/ToF) registration program (C++, Qt, VTK).

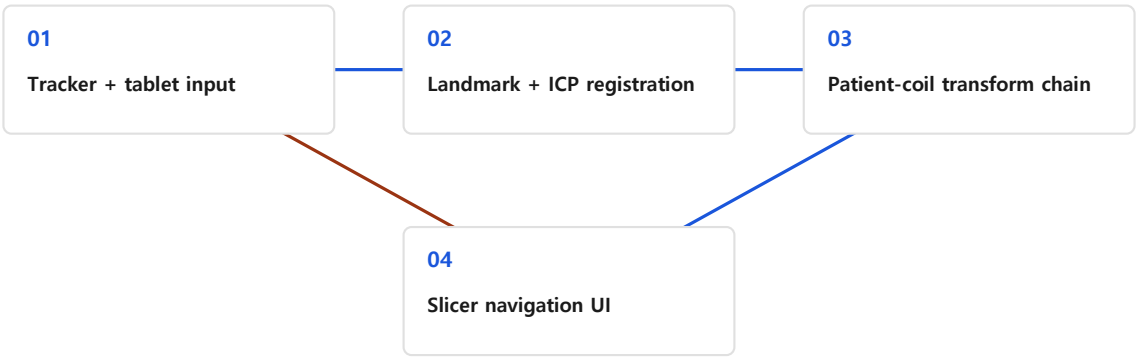
Navigation UI workflow

Six Home-screen menus gate the run from DICOM load through reconstruction, registration, and targets to the live session, with crosshair aiming.

Coordinate and registration engine

Image, patient, tracker, and coil transforms stay explicit; ICP offers three modes.

Coordinate loop from input to aiming guidance



Explanatory system relationship diagram; it is not photographic or experimental evidence.

Verification and evidence ledger

The evidence is the license-gated deployment build, a regression-test suite grown issue by issue, and

coordinate-transform and registration logs.

Device and system integration

Implemented the tracker SDK migration, tablet TCP protocol, and robot interfaces. Moved the optical-tracking SDK from v6 to v7.1.2 and connected customer-provided AI modules through an adapter boundary updated by swapping a folder. Integration discussions in 2026 consolidated registration in NeuroPilot and separated target-file, pose-data, and synchronization requirements. Programme participation began in July 2024; the main NeuroPilot repository implementation is recorded from July 2025.

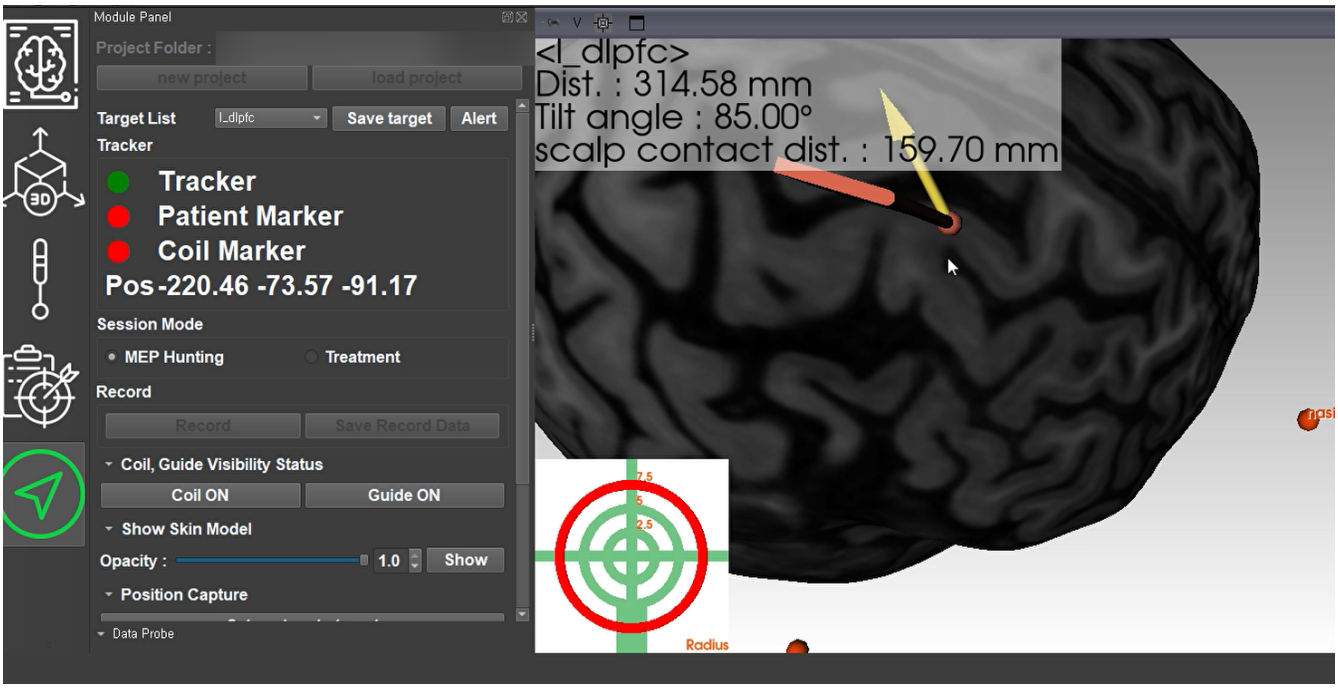


Figure 2. NeuroPilot navigation module: tracker, patient-marker and coil-marker status with distance and tilt guidance to the target

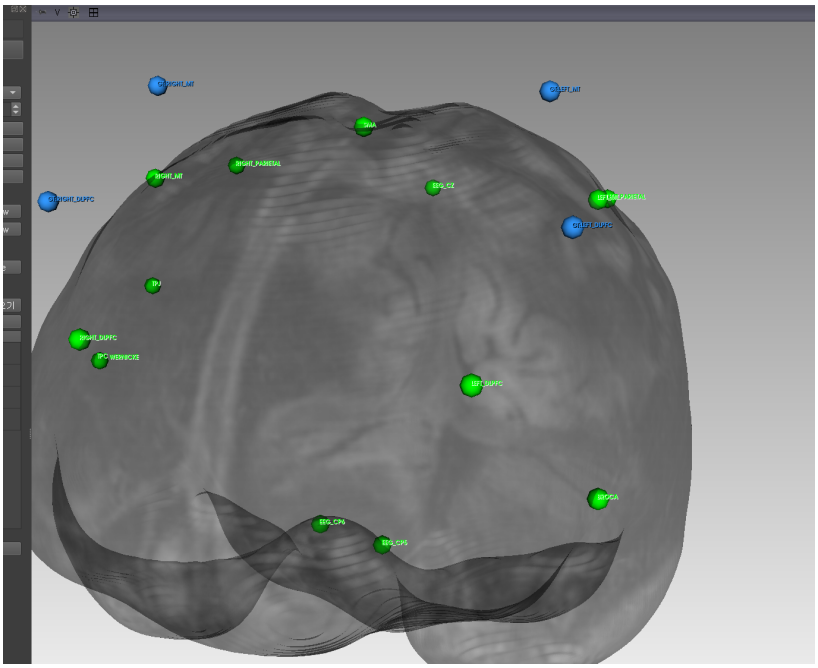


Figure 3. NeuroPilot target view: stimulation targets labelled on the cortical surface model

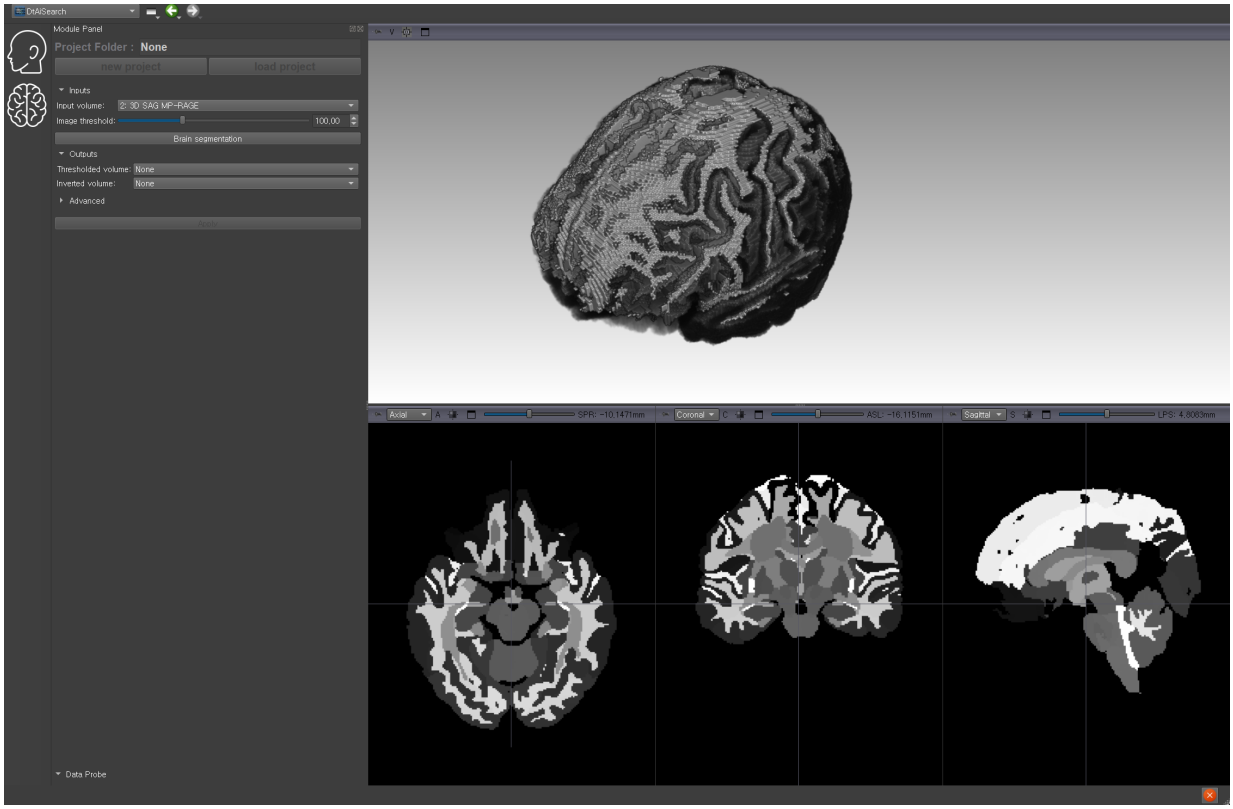


Figure 4. AI brain segmentation: a skull-stripped 3D brain model with tissue segmentation on the axial, coronal, and sagittal slices

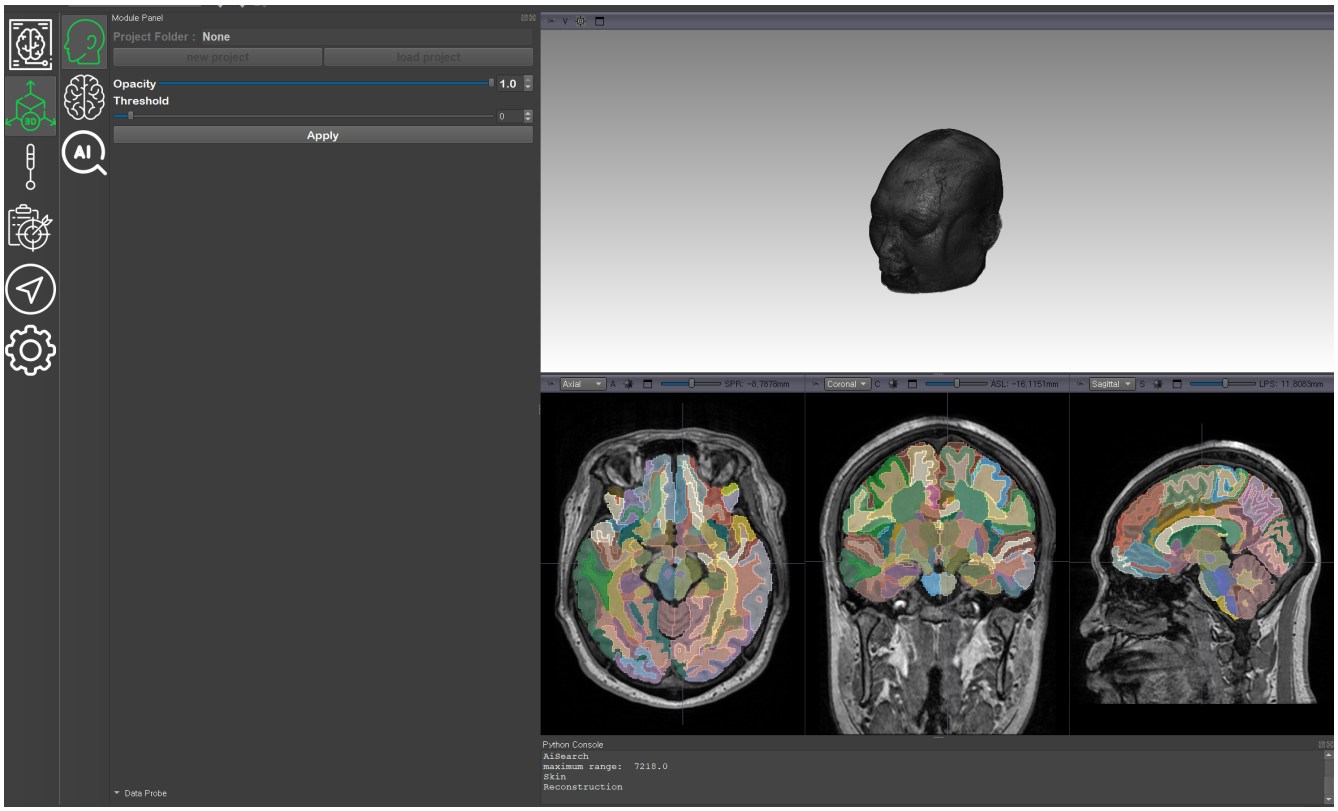


Figure 5. Cortical parcellation: region-coded segmentation overlaid on the slice views, with the reconstructed head surface

Role, team result, and limits

Team result

AT&C owns license operations and adoption. I designed the core structure; a colleague built specific screens.

Limitations

No clinical efficacy, quantitative accuracy, or regulatory status is claimed, and patient data stays out of this case.

Clinical and product boundary

By 1 June 2026 I had implemented reference-coordinate loading, 3D markers, per-target distance and RMSE calculation, and CSV export. Sample review exposed anatomical-region mapping and reference-coordinate quality issues, so I requested target definitions and remeasurement under a protocol still to be agreed. The quantitative-evaluation screen is kept out of the delivered build and kept as a tool in a separate evaluation branch; after the August 2026 handover I take part as an adviser. Implementing the comparison tool does not establish a valid final evaluation; no clinical efficacy, quantitative accuracy, or regulatory status is claimed.

Collaboration boundary

Integrated customer-provided AI models into the application and agreed on registration, target, and communication responsibilities and evaluation-data definitions. Reference-coordinate quality issues led to a remeasurement request; model training and development remain separate responsibilities.

Implementation technologies

3D Slicer · VTK · PyQt5 · Python · PyTorch (inference integration) · Optical tracking · TCP/IP binary protocol · License gating

Primary capability

Surgical Navigation & Optical Tracking · 3D Registration & Computer Vision

Registered public evidence

No external public link is currently registered.

What to bring

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

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